

Shift11

AI-Powered Football Momentum Attribution

Theoretical Project & System Design

Problem Statement

Traditional football statistics describe individual events but often fail to explain the turning point of a match: when momentum changed, which players drove the change, what sequence caused it, and why the change mattered tactically. Shift11 addresses this gap by connecting deterministic event analytics, player attribution, evidence-chain reconstruction and grounded AI explanation.

Proposed Solution / Logic

Stage	Logic
1. Ingest	Load validated JSON events containing timestamp, team, player, event type, location, outcome and context.
2. Match State	Maintain territory, progression, pressure/recoveries, chance creation and possession-control features.
3. Momentum	Normalize five components to 0–100 and weight them: territory 20%, progression 25%, pressure/recoveries 20%, chance creation 25%, possession control 10%.
4. Window	Use a 5-minute rolling window updated every 30 seconds; recent events receive greater weight.
5. Shift	Trigger at momentum increase ≥ 15 , opponent-advantage increase ≥ 10 , persistence ≥ 60 seconds, and ≥ 3 meaningful supporting events. Hysteresis/cooldown prevents repeated markers.
6. Attribution	Score player contributions from event relevance and sequence position; rank primary/secondary contributors and assign tactical roles.
7. Evidence	Reconstruct a coherent 5–7 event chain directly supporting the shift.
8. Fact Pack	Package verified shift metrics, contributors, roles, confidence and evidence.
9. AI	Send only the fact pack to Agent Framework/Foundry; require grounded explanation and prohibit unsupported facts.
10. UI	Render timeline markers, shift details, player attribution, evidence, before/after metrics and explanation.

System Architecture

Synthetic Event Generator → Event Validation → Match State → Momentum Engine → Shift Detector → Player Attribution → Evidence Chain / Fact Pack → Microsoft Agent Framework → Microsoft Foundry → Interactive Demo UI

Tools & Technologies

Tool	Purpose
Python	Core implementation, event generation and deterministic analytics.
Pandas / NumPy (optional)	Rolling-window calculations and feature engineering where useful.
JSON + validation	Event and fact-pack representation; validation keeps malformed events out.
Microsoft Agent Framework	Agent/orchestration layer for grounded tactical explanations.
Microsoft Foundry	Microsoft AI/model/runtime path and potential hosted-agent deployment.
Azure	Deployment and supporting cloud services where useful.
Git + GitHub / GitHub Desktop	Version control and reproducible development workflow.
Interactive UI/API stack	Timeline, shift drill-down and evidence presentation.

Relational Schema (Theoretical / Optional)

The MVP can operate entirely on JSON. A relational database becomes useful for persistent multi-match history and querying.

